

Mathematical Induction (গাণিতিক আৱেষ্ণা)

26/9/23

CSE-230

* It is a way to prove a mathematical problem.

prove,

$$1+2+3+4+\dots+(n-1)+n = \frac{n(n+1)}{2} \quad [n \in \mathbb{N}]$$

$$1+2+3+4+\dots+(n-1)+n = \frac{n(n+1)}{2} \quad \text{--- (i)}$$

Step-1: for $n=1$

$$\text{L.H.S} = 1$$

$$\text{R.H.S} = \frac{1 \times (1+1)}{2} = 1 = \text{L.H.S}$$

Step-2:

let's assume that for $n=m$, (i) is true.

$$\Rightarrow 1+2+3+4+\dots+(m-1)+m = \frac{m(m+1)}{2} \quad \text{--- (ii)}$$

Step-3:

for $n=m+1$

$$\text{L.H.S} = 1+2+3+4+\dots+m+(m+1)$$
$$= \frac{m(m+1)}{2} \quad [\text{from ii}] + (m+1)$$

$$= \frac{m(m+1) + 2(m+1)}{2}$$

$$= \frac{(m+1)(m+2)}{2}$$

$$= \text{R.H.S}$$

for, $n=m+1$

$$\text{R.H.S} = \frac{(m+1)(m+2)}{2}$$

Therefore, for all $n \in \mathbb{N}$, (i) is true.
 [proved]

Prove that, for all $n \in \mathbb{N}$

$$1^2 + 2^2 + 3^2 + \dots + (n-1)^2 + n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$1^2 + 2^2 + 3^2 + \dots + (n-1)^2 + n^2 = \frac{n(n+1)(2n+1)}{6} \quad \text{--- (i)}$$

Step-1:

$n=1$

$$\text{L.H.S} = 1^2 = 1$$

R.H.S

$$= \frac{1(1+1)(2 \times 1+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1 = \text{L.H.S}$$

Step-2:

Let's assume that for $n=m$, (i) is true.

$$\Rightarrow 1^2 + 2^2 + 3^2 + \dots + (m-1)^2 + m^2 = \frac{m(m+1)(2m+1)}{6} \quad \text{--- (ii)}$$

Step-3:

For, $n = m+1$

$$\text{L.H.S} = 1^2 + 2^2 + 3^2 + \dots + m^2 + (m+1)^2$$

$$= \frac{m(m+1)(2m+1)}{6} + (m+1)^2$$

$$= \frac{m(m+1)(2m+1) + 6(m+1)^2}{6}$$

$$= \frac{(m+1) \{ m(2m+1) + 6(m+1) \}}{6}$$

$$= \frac{(m+1)(2m^2 + m + 6m + 6)}{6}$$

$$= \frac{(m+1)(2m^2 + 7m + 6)}{6}$$

For, $n = m+1$,

$$\text{R.H.S} = \frac{(m+1)(m+2) \{ 2(m+1) + 1 \}}{6}$$

$$= \frac{(m+1)(m+2)(2m+3)}{6}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

$$\frac{(m+1) \{ 2m^2 + 7m + 6 \}}{6} = \frac{(m+1)(m+2)(2m+3)}{6}$$

$$= \frac{(m+1)(m+2)(2m+3)}{6}$$

$$= \text{R.H.S}$$

□ Prove that,

$$1^3 + 2^3 + 3^3 + \dots + (n+1)^3 + n^3 = \left\{ \frac{n(n+1)}{2} \right\}^2$$

L.H.S:

$$1^3 = 1$$

R.H.S:

$$\left\{ \frac{1(1+1)}{2} \right\}^2 = 1^2 = 1$$

$n = m$

$$1^3 + 2^3 + 3^3 + \dots + (m+1)^3 + m^3 = \left\{ \frac{m(m+1)}{2} \right\}^2$$

$n = m+1$

$$1^3 + 2^3 + 3^3 + \dots + (m+1)^3 + (m+1)^3$$

$$= \left\{ \frac{m(m+1)}{2} \right\}^2 + (m+1)^3$$

$$= \left\{ \frac{m(m+1)}{2} \right\}^2 + (m+1)^3$$

$$= \frac{m^2(m+1)^2}{4} + (m+1)^3$$

$$= \frac{m^2(m+1)^2 + 4(m+1)^3}{4}$$

$$= \frac{(m+1)^2 (m^2 + 4m + 4)}{4}$$

$$= \frac{(m+1)^2 (m+2)^2}{4}$$

$$= \left\{ \frac{(m+1)(m+2)}{2} \right\}^2$$

show that, the sum of first n odd positive integer is n^2

step-1 →

$$1+3+5+7+\dots+(2n-3)+(2n-1)=n^2 \quad \text{--- (i)}$$

Let $n=1$

L.H.S = 1

R.H.S

$$n^2 = 1^2 = 1 = \text{L.H.S}$$

step-2 →

Let, $n=m$

$$1+3+5+7+\dots+(2m-3)+(2m-1)=m^2 \quad \text{--- (ii)}$$

step-3 →

$$n=m+1$$

L.H.S:

$$\begin{aligned} &1+3+5+7+\dots+(2(m+1)-3)+(2(m+1)-1) \\ &= 1+3+5+7+\dots+(2m+2-3)+(2m+2-1) \\ &= 1+3+5+7+\dots+(2m-1)+(2m+1) \end{aligned}$$

$$= m^2 + (2m+1)$$

$$= (m+1)^2$$

$$= \boxed{\text{R.H.S}}$$

$$\text{R.H.S} = (m+1)^2$$

Prove that for all $n \in \mathbb{N}$

$$2^n \geq n$$

$$2^n > n \quad \text{--- (i)}$$

Step-1

$$n=1 \quad \text{L.H.S} = 2^1 = 2$$

$$\text{R.H.S} = 1$$

$$n \leq 1$$

$$2 > 1$$

$$\boxed{\text{L.H.S} > \text{R.H.S}}$$

Step-2

... for $n=m$, (i) is true $\Rightarrow 2^m > m$

$$2^m > m$$

$$\Rightarrow 2 \times 2^m > 2m \Rightarrow 2^{m+1} > 2m \quad \text{--- (ii)}$$

$$\text{(ii)} - (1+m)m \leq (1+m) \times m$$

Step-3

for $n=m+1$

$$\text{L.H.S} = 2^{m+1} > 2m$$

$$\text{R.H.S,} \\ = (m+1)$$

$$\boxed{\begin{matrix} m \geq 1 \\ m+m \geq m+1 \end{matrix}}$$

$$= m+m \geq m+1$$

$$= \text{R.H.S}$$

$$\boxed{\begin{matrix} (1+m) \leq (1+m)m \\ \therefore \text{L.H.S} > \text{R.H.S} \end{matrix}}$$

$$(1+m) \leq (1+m)$$

$$2.H.9 \leq 2.H.1$$

Prove that,

for $n \in \mathbb{N}$

$$n! \geq n$$

Step-1:

$$n = 1$$

$$\text{L.H.S} = 1! = 1$$

$$\text{R.H.S} = 1$$

$$\therefore \text{L.H.S} \geq \text{R.H.S}$$

Step-2:

$$n = m$$

$$m! \geq m$$

$$m \times (m+1)! \geq m(m+1) \quad \text{--- (ii)}$$

Step-3:

$$n = m+1$$

$$\text{L.H.S} = (m+1)!$$

$$= m! (m+1)$$

$$\geq m(m+1)$$

$$\geq m+1$$

$$= \text{R.H.S}$$

$$\text{L.H.S} \geq \text{R.H.S}$$

$$(m+1)! \geq (m+1)$$

$$\boxed{\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}} \quad \text{--- (i)}$$

Step-1 →

for $n = 1$,

$$\begin{aligned} \text{L.H.S} &= \frac{1}{1 \times (1+1)} \\ &= \frac{1}{2} \end{aligned}$$

$$\begin{aligned} \text{R.H.S} &= \frac{1}{1+1} = \frac{1}{2} \\ &= \frac{1}{2} \end{aligned}$$

Step-2 →

for $n = m$

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{m(m+1)} = \frac{m}{m+1} \quad \text{--- (ii)}$$

Step-3 →

$n = m+1$

$$\boxed{\text{R.H.S: } \frac{m+1}{(m+1)(m+2)}}$$

L.H.S →

$$\begin{aligned} &\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{m(m+1)} + \frac{1}{(m+1)(m+2)} \\ &= \frac{m}{m+1} + \frac{1}{(m+1)(m+2)} \\ &= \frac{m(m+2) + 1}{(m+1)(m+2)} \\ &= \frac{m^2 + 2m + 1}{(m+1)^2(m+2)} = \frac{(m+1)^2}{(m+1)^2(m+2)} \\ &= \frac{m+1}{m+2} \end{aligned}$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

□ Show ~~that~~ by mathematical Induction that $19^n - 10^n$ is always divisible by 9 for all $n \in \mathbb{N}$

Step-1 → for $n = 1$
 $19^1 - 10^1 = 9$; and 9 is surely divisible by his own number.

Step-2 → for $n = m$

$$19^m - 10^m = 9k \quad [k \in \mathbb{N}]$$

$$\Rightarrow 19^m = 9k + 10^m \quad \text{--- (i)}$$

Step-3 → $n = m+1$

$$\begin{aligned} & 19^{m+1} - 10^{m+1} \\ &= 19 \cdot 19^m - 10^{m+1} \\ &= 19(9k + 10^m) - 10^{m+1} \\ &= 19(9k + 10^m) - 10^m \cdot 10 \\ &= 19 \cdot 9k + 19 \cdot 10^m - 10^m \cdot 10 \\ &= 19 \cdot 9k + 10^m(19 - 10) \\ &= 19 \cdot 9k + 9 \cdot 10^m \\ &= 9(19k + 10^m) \end{aligned}$$





$2^n \times 2^n$ size এর কোন Board-এ একটি cell না থাকলে, এই domino বসানোকে overlap না করে পুরোটা black করে ফেলা যায়?

Step-1

for $n=1$,
 $2^1 \times 2^1$

$2^n \times 2^n \rightarrow 1$ cell eaten

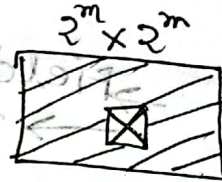


Step-2

$n=m$,

$2^m \times 2^m$

fully black

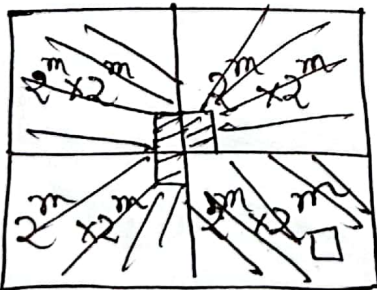


Step-3

$n=m+1$

$$2^{m+1} \times 2^{m+1} = 2 \cdot 2^m \times 2 \cdot 2^m$$

$$2^{m+1} \times 2^{m+1} = 4 \times (2^m \times 2^m)$$



Number Theory Related Induction Problem

$$\boxed{\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{n(n+1)}} = \frac{n}{n(n+1)}$$

Step-1 $\rightarrow$
 $n=1$

L.H.S $\rightarrow$

$$\frac{1}{1(1+1)} = \frac{1}{2}$$

R.H.S $\rightarrow$

$$\frac{1}{1+2} = \frac{1}{3}$$

Step-2 $\rightarrow$

$$\frac{1}{m(m+1)} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{m(m+1)} = \frac{m}{m(m+1)}$$

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Step-3 $\rightarrow$ $n=m+1$

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(m+1)(m+2)}$$

$$= \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{m^2 + 3m + 3}$$

$\rightarrow$ R.H.S = $\frac{m+1}{(m+1)(m+2)}$

Step-3

$$n = m + 1$$

$$\text{L.H.S} = \left\{ \frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{m(m+1)} \right\} + \frac{1}{(m+1)(m+2)}$$

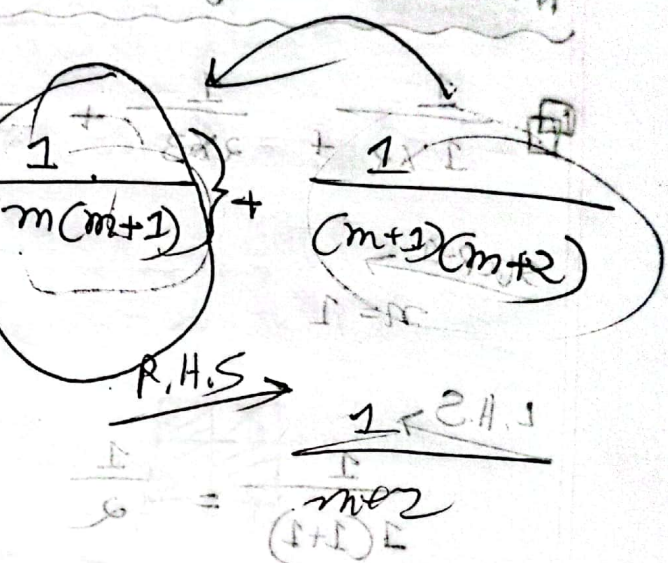
$$= \frac{m}{m+1} + \frac{1}{(m+1)(m+2)}$$

$$= \frac{m(m+2) + 1}{(m+1)(m+2)}$$

$$= \frac{m^2 + 2m + 1}{(m+1)(m+2)^2}$$

$$= \frac{1}{m+1} + \frac{1}{(m+1)(m+2)} + \dots + \frac{1}{(m+1)(m+2)}$$

$$\text{R.H.S}$$



$$\frac{1}{(m+1)(m+2)^2}$$

$$m+1 = n$$

$$\frac{1}{(m+1)(m+2)} + \dots + \frac{1}{(m+1)(m+2)}$$

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \dots + \frac{1}{m \times (m+1)} =$$

=

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$$10 = 2 \times 5 + 0$$

ভাজক = ভাজক $\times$ ভাজ্য + ভাজ্য

Dividend = Divisor $\times$ Quotient + Remainder

divisor $\leftarrow$ ১২ (২ $\rightarrow$ quotient
 $\frac{10}{2} = 2.H.I$
 ২ $\rightarrow$ Remainder

Show that (Induction),

$19^n - 10^n$ is always divisible by 9 for all $n \in \mathbb{N}$

Step-1 $\rightarrow n=1$

$19^1 - 10^1 = 9K$ $\rightarrow 19^m = 9K + 10^m$ (ii)

Step-2

For $n=m$, $19^m - 10^m = 9K$ [K is an $\mathbb{N}$]

Step-3

$n=m+1$

$$19^{m+1} - 10^{m+1}$$

$$= 19^1 \times 19^m - 10^{m+1}$$

$$= 19(9K + 10^m) - 10^{m+1}$$

$$= 19 \cdot 9K + 19 \cdot 10^m - 10^{m+1}$$

$$= 19 \cdot 9K + 10^m (19 - 10)$$

$$= 19 \cdot 9K + 10^m \times 9$$

$$= 9(19K + 10^m)$$

for $n=m+1 \rightarrow$

$9(19K + 10^m)$ is divisible by 9 so any

$n \in \mathbb{N}$



$$x^2 = 8P + 1$$

$$m^2 = 8P + 1$$

$$n = m + 1$$

$$= 1$$



$$1 + 3 + 5 + 7 + \dots + (2n-3) + (2n-1) = n^2$$

$$n=1$$

$$L.H.S = 1$$

$$R.H.S = 1$$

$$n=m$$

$$1 + 3 + 5 + \dots + (2m-3) + (2m-1) = m^2$$

$$n=m+1$$

$$1 + 3 + 5 + \dots + (2(m+1)-3) + (2(m+1)-1) = (m+1)^2$$

$$= 1 + 3 + 5 + \dots + (2m-1) + (2m+1)$$

$$= m^2 + 2m + 1$$

$$= (m+1)^2$$

□ State that, $n^2 - 3n + 4$ is always even for $n \in \mathbb{N}$

step-1 →

step-2 →

$$m^2 - 3m + 4 = 2K \quad (\text{Assumption})$$

step-3 →

$$\begin{aligned} & (m+1)^2 - 3(m+1) + 4 \\ &= (m+1)^2 - 3(m+1) + 4 \\ &= m^2 + 2m + 1 - 3m - 3 + 4 \\ &= 2K + 2m - 2 \\ &= 2(K + m - 1) \end{aligned}$$

Alternative for $n \in \mathbb{N}$

$$\begin{aligned} & n^2 - 3n + 4 \\ &= n^2 - 3n + 2 + 2 \\ &= n^2 - 2n - n + 2 + 2 \\ &= n(n-2) - 1(n-2) + 2 \\ &= (n-2)(n-1) + 2 \\ &= 2K + 2 \end{aligned}$$

$$\square \quad \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \frac{1}{7 \times 9} + \dots + \frac{1}{(2n+1)(2n+3)} = \frac{n}{3(2n+3)} \quad (i)$$

S-1 → $n=1$

$$\text{L.H.S.} = \frac{1}{15}$$

$$\text{R.H.S.} = \frac{1}{15}$$

S-2 →

$$S-3 \rightarrow \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2m+1)(2m+3)} + \frac{1}{(2m+3)(2m+5)}$$

$$= \frac{m}{3(2m+3)} + \frac{1}{(2m+3)(2m+5)}$$

$$= \frac{(m+1)(2m+3)}{3(2m+3)(2m+5)}$$

$$= \frac{m+1}{3(2m+5)}$$

$$A + (1+m)E - S(1+m) =$$

$$P + (1+m)E - S(1+m) =$$

$$mS + Sm + 1 - Sm - S + 1 =$$

$$S - mS + 2 =$$

$$S(1-m+2) =$$

$$\frac{1}{E} = \frac{1}{(S+m)(S+m)} + \dots + \frac{1}{E \times F} + \frac{1}{F \times E} + \frac{1}{E \times E}$$

$$\frac{1}{E} = 2.4.9$$

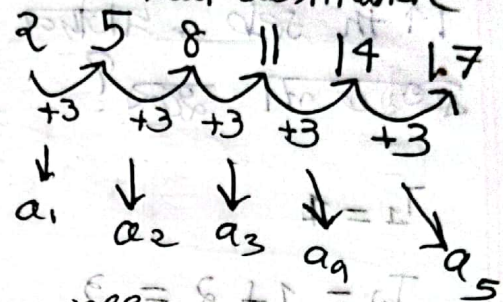
$$E = 1 \rightarrow \frac{1}{E} = 2.4.9$$

Recurrence (ধনবর্ধিত)

method $\rightarrow$ Backtracking

Back substitution

Step-1: Boundary condition apply



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$$a_1 = 2$$

Step-2: Recurrence Relation

$$a_n = a_{n-1} + 3$$

কোন একটি sequence

জোরজোড়ের ক্ষেত্রে টাইপ
সামান্য বিজ্ঞান Related

$$a_{23} = ?$$

Step-3: solution to the Recurrence

(n সং function এর সাথে
Related করা)

$$a_2 = a_1 + 3$$

$$a_3 = a_2 + 3$$

$$a_4 = a_3 + 3$$

$$a_n = a_{n-1} + 3 \times 1$$

$$= (a_{n-2} + 3) + 3$$

$$= (a_{n-3} + 3) + 3 \times 2$$

$$= (a_{n-3} + 3) + 3 \times 2$$

$$= a_{n-3} + 3 \times 3$$

$$= (a_{n-4} + 3) + 3 \times 3$$

$$= a_{n-4} + 3 \times 4$$

$$= a_{n-(n-1)} + 3 \times (n-1)$$

$$= a_1 + 3(n-1)$$

$$= 2 + 3n - 3 \quad a_n = 3n - 1$$

$$r^n = 8^{n-1}$$

$$r = 1$$

$$(a_n)^p = A_0$$

$$A_0 = A_0 + 3$$

$$a_{2023} = 3 \times 2023 - 1$$

$$a_{2023} = 6068$$

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২য় অর্ধ (২) নং

প্রথমটি বুঝে গেলে বাকি বাকি হলো, 1st Jan 14th Feb. প্রতিদিন double করা হলো এ কয়দিনে হওয়া লাগবে?

$$T_1 = 1$$

Step 1

$$T_1 = 1$$

$$T_2 = 1 + 2 = 3$$

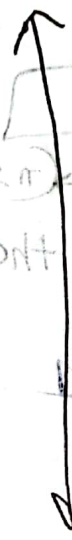
$$T_3 = 3 + 3 \times 2 = 9$$

Step 2

$$T_n = T_{n-1} + 2 \times T_{n-1}$$

Step 3

$$E + T_n = 3T_{n-1}$$



$$r^n = 3r^{n-1}$$

$$r^1 = 3$$

$$a_n = a_1 r^n$$

$$n=1, 1 = a_1 / 3 \Rightarrow a_1 = 3$$

$$T_n = \frac{1}{3} \times 3^n$$

(Ans)

$$E + (E + 3^n) =$$

$$= 3 \times (3 \times T_{n-2})$$

$$= 3^2 \times T_{n-2}$$

$$= 3^2 \times (3 \times T_{n-3})$$

$$= 3^3 \times T_{n-3}$$

$$= 3^4 \times T_{n-4}$$

$$= 3^5 \times T_{n-5}$$

$$\vdots$$

$$= 3^{n-1} \times T_{n-(n-1)}$$

$$T_n = 3^{n-1} \times T_1$$

$$T_n = 3^{n-1}$$

$$T_{15} = 3^{14}$$

$$= 9.8972 \times 10^8$$

$$(1-n) \times E + (1-n) \times 0 =$$

$$-nE = 0 - nE + 0$$

Recursion →

500 tk
 2024 January → 2% cumulative
 maximum interest limit

and After the month government cuts 2tk.

So, 2025 August कब तक?

$$a_0 = 500$$

$$a_1 = a_0 + 2\% a_0 - 2$$

$$a_n = a_{n-1} + 2\% a_{n-1} - 2$$

$$a_n = a_{n-1} + 0.02 a_{n-1} - 2$$

$$a_n = 1.02 [a_{n-1} - 2]$$

$$= 1.02 (1.02 a_{n-2} - 2) - 2$$

$$a_n = 1.02^2 [a_{n-2} - 2 \times 1.02 - 2]$$

$$= 1.02^2 (1.02 a_{n-3} - 2) - 2 \times 1.02 - 2$$

$$a_n = 1.02^3 (1.02 a_{n-4} - 2) - 2 \times 1.02^2 - 2 \times 1.02 - 2$$

$$= 1.02^4 a_{n-4} - 2 \times 1.02^3 - 2 \times 1.02^2 - 2 \times 1.02 - 2$$

$$\frac{a(n-1)}{(r-1)}$$

$$2 \xrightarrow{3} 9 \xleftarrow{3} 8$$

बुनाडा
दीया

$$\frac{a(n-1)}{n-1}$$

$$a_n = 1.02^n a_{n-n} - 2 \times 1.02^{n-1} - 2 \times 1.02^{n-2} - 2 \times \dots - 2 \times 1.02 - 2$$

$$a_n = 1.02^n \times a_0 - 2(1.02^{n-1} + 1.02^{n-2} + 1.02^{n-3} + \dots + 1)$$

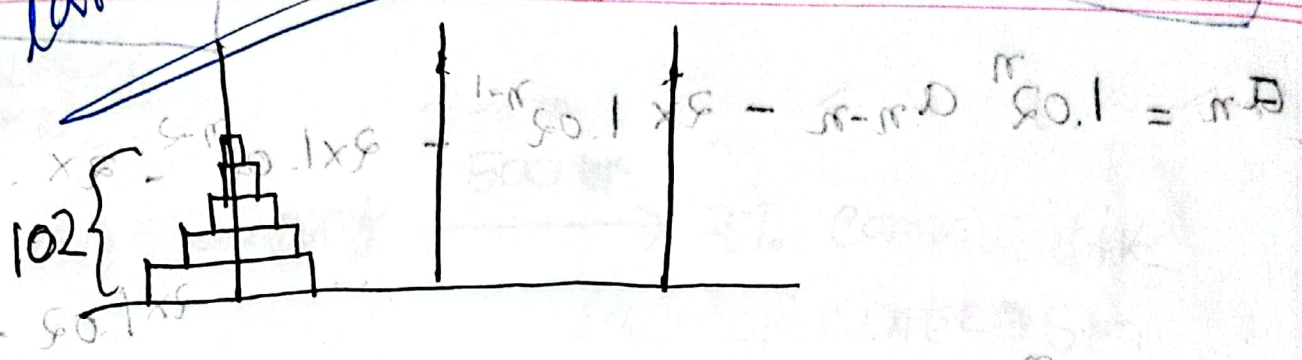
$$a_n = 1.02^n \times 500 - 2 \times \frac{1 \times (1.02^n - 1)}{(1.02 - 1)}$$

$$= 1.02^n \times 500 - 2 \times \frac{1.02^n - 1}{0.2}$$

$$= 1.02^n \times 500 - 100(1.02^n - 1)$$

$$a_n = 400 \times 1.02^n + 100$$

last member of end



$$T_n = T_{n-1} + 1 + T_{n-1}$$

$$T_n = 1.2 T_{n-1} + 1$$

TBC

Linear
Recurrence

non homogeneous

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ T_n &= 3T_{n-1} \\ a_n &= 1.02a_{n-1} \\ T_n &= 2T_{n-1} + 1 \end{aligned}$$

Solution of a linear Homogeneous Recurrence (LHR)

$$a_n = C_1 a_{n-1} + C_2 a_{n-2} + C_3 a_{n-3} + \dots + C_k a_{n-k}$$

$$r^n = C_1 r^{n-1} + C_2 r^{n-2} + C_3 r^{n-3} + \dots + C_k r^{n-k}$$

$$r^k = C_1 r^{k-1} + C_2 r^{k-2} + C_3 r^{k-3} + \dots + C_k r^0$$

$$r^k = C_1 r^{k-1} + C_2 r^{k-2} + \dots + C_k \rightarrow \text{characteristic equation}$$

$$C_1 r_1, r_2, r_3, \dots, r_m$$

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \alpha_3 r_3^n + \dots + \alpha_m r_m^n$$

⑤ ①, ii $\rightarrow$ ①

$$a_n = 4^n - 3^n$$

$n=1,$

$$a_1 = \alpha_1 \times 4^1 + \alpha_2 \times 8$$

$$1 = 4\alpha_1 + 8\alpha_2 \quad \text{--- ②}$$

④ solving ① and ②

$$\alpha_1 = 1$$

$$\alpha_2 = -1$$

$$\rightarrow a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

$$a_0 = \alpha_1 \times 4^0 + \alpha_2 \times 3^0 \quad \text{--- ①}$$

$$\begin{aligned} a_0 &= 0 \\ a_1 &= 1 \end{aligned}$$

$$\text{①} \leftarrow a_n = 7a_{n-1} - 12a_{n-2}$$

$$r^n = 7r^{n-1} - 12r^{n-2}$$

$$r^2 = 7r - 12$$

$$\begin{aligned} r_1 &= 4 \\ r_2 &= 3 \end{aligned}$$

$$\Rightarrow r^2 - 7r + 12 = 0$$

10/10/23

(9H) $\boxed{a_0 = 0}$
 $a_1 = 2$

$$a_{n+2} = 9a_{n+1} - 20a_n$$

$$\Rightarrow r^{n+2} = 9r^{n+1} - 20r^n$$

$$\Rightarrow r^2 = 9r - 20 \quad [\text{dividing by } r^n]$$

$$\Rightarrow r^2 - 9r + 20 = 0$$

$$\Rightarrow r^2 - 4r - 5r + 20 = 0$$

$$\Rightarrow (r-4)(r-5) = 0$$

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n$$

$$a_n = \alpha_1 (4)^n + \alpha_2 (5)^n$$

$$n=0 \Rightarrow$$

$$a_0 = \alpha_1 4^0 + \alpha_2 5^0$$

$$\alpha_1 + \alpha_2 = 0 \quad \text{--- (ii)}$$

--- (i)

$$n=1 \Rightarrow$$

$$a_1 = \alpha_1 4^1 + \alpha_2 5^1$$

$$2 = 4\alpha_1 + 5\alpha_2$$

⑦) x4 - ③)

$$0 - 2 = 4\alpha_1 + 4\alpha_2 - 4\alpha_1 - 5\alpha_2$$

$$-2 = -\alpha_2 \Rightarrow \alpha_2 = -2$$

$$\alpha_1 = 2$$

putting the value $\alpha_1 = 2, \alpha_2 = -2$ in equation ①

$$a_n = -2 \times 4^n + 2 \cdot 5^n$$

$$\square a_0 = 0$$

$$a_1 = 2$$

$$a_2 = 4$$

$$a_{n+3} = 9a_{n+1} - 3a_{n+2} - 5a_n$$

$$\Rightarrow r^{n+3} = 9r^{n+1} - 3r^{n+2} - 5r^n$$

$$\Rightarrow r^3 = 9r - 3r^2 - 5$$

$$\Rightarrow r^3 + 3r^2 - 9r + 5 = 0$$

$$\Rightarrow r^2(r-1) + 4r(r-1) - 5(r-1) = 0$$

$$\Rightarrow (r-1)(r^2 + 4r - 5) = 0$$

$$\Rightarrow (r-1)\{r^2 - r + 5r - 5\} = 0$$

$$\Rightarrow (r-1)\{r(r-1) + 5(r-1)\} = 0$$

$$\Rightarrow (r-1)(r-1)(r+5)=0$$

$$r_1=1$$

$$r_2=1$$

$$r_3=-5$$

its roots are same,

$$a_n = \alpha_1 r_1^n + \alpha_2 r_2^n + \alpha_3 r_3^n$$

$$a_n = \alpha_1 \cdot 1^n + \alpha_2 (1)^n + \alpha_3 (-5)^n$$

$$a_n = \alpha_1 + \alpha_2 n + \alpha_3 (-5)^n \quad \text{--- (i)}$$

$$n=0 \Rightarrow a_0 = \alpha_1 + \alpha_2 \times 0 + \alpha_3 (-5)^0$$

$$0 = \alpha_1 + \alpha_3 \quad \text{--- (ii)}$$

$$n=1 \Rightarrow$$

$$a_1 = \alpha_1 + \alpha_2 \times 1 + \alpha_3 (-5)^1$$

$$2 = \alpha_1 + \alpha_2 - 5\alpha_3 \quad \text{--- (iii)}$$

(iii) - Ans

$$S = A + A^2 + A^3 + \dots + A^n = S$$

$$S = A + A^2 + \dots + A^n$$

$$S = A + A^2 + \dots + A^n$$

$$S = A + A^2 + \dots + A^n$$

$$0 = 0$$

$$S = 1$$

$$A = 1$$

$$n=2 \Rightarrow$$

$$a_2 = \alpha_1 + \alpha_2 \times 2 + \alpha_3 (-5)^2$$

$$\Rightarrow 4 = \alpha_1 + 2\alpha_2 + 25\alpha_3 \quad \text{--- (iv)}$$

solving (i), (ii) and (iii)

$$\alpha_1 = 0, \alpha_2 = 2, \alpha_3 = 0$$

Putting the value of $\alpha_1, \alpha_2, \alpha_3$ in eqn (I) $\Rightarrow$

$$a_n = 0 + 2 \times n + 0 \times (-5)^2$$

$$\boxed{a_n = 2n}$$

Solving linear non-homogeneous recurrence

$$\boxed{a_0=2, a_1=9} \\ a_n = 2a_{n-1} + 3a_{n-2} + 5^n$$

just indicate not solve

Homogeneous (h)

(1)

$$a_n = (a_n)^h + (a_n)^p \quad \text{--- (511)}$$

$$\begin{aligned} &+1 \\ &+2n \\ &+n^2 \\ &+125n^3 \end{aligned}$$

Particular Part (p)

$$a_n^h = 2a_{n-1} + 3a_{n-2}$$

$$\Rightarrow r^n = 2r^{n-1} + 3r^{n-2}$$

$$\Rightarrow r^2 = 2r + 3$$

$$\Rightarrow r^2 - 2r - 3 = 0$$

$$\Rightarrow (r-3)(r+1) = 0$$

$$a_n^h = \alpha_1 r_1^n + \alpha_2 r_2^n$$

$$\Rightarrow a_n^h = \alpha_1 3^n + \alpha_2 (-1)^n$$

$$\boxed{r_1 = 3}$$

$$\boxed{r_2 = -1}$$

Particulars part

$$\textcircled{1} \leftarrow C$$

$$\textcircled{2} \leftarrow n$$

$$25n^2 \leftarrow n^2$$

$$125n^3 \leftarrow n^3$$

$$5^n \leftarrow b^{n \times \lambda}$$

Assumed

Sol. of P.P.

$$A_0$$

$$A_1 n + A_0$$

$$A_2 n^2 + A_1 n + A_0$$

$$A_3 n^3 + A_2 n^2 + A_1 n + A_0$$

$$A b^n$$

$$a_n^p = A b^n = A \times 5^n \quad \text{--- (iii)}$$

~~(iii)~~ (iii) $\rightarrow$ (i)

$$A \times 5^n = 2 \times A \times 5^{n-1} + 3 \times A \times 5^{n-2} + 5^n \times 1 = 5^n$$

$$\Rightarrow A \times 5^2 = 2 \times A \times 5 + 3 \times A + 5^2 = \boxed{5^{n-2} \text{ दिया जाय}} \quad \text{--- (1)}$$

$$\Rightarrow 25A = 10A + 3A + 25$$

$$\Rightarrow \boxed{A = \frac{25}{12}}$$

$$a_n^p = A \times 5^n = \frac{25}{12} \times 5^n$$

$$= \frac{5^2}{12} \times 5^n$$

$$= \frac{5^{n+2}}{12} \quad \text{--- (iv)}$$

(ii) (iv) $\rightarrow$ (iv)

$$L \times P = \frac{51}{5}$$

$$a_n = \alpha_1 3^n + \alpha_2 (-1)^n + \frac{5^{n+2}}{12}$$

ANSWER

$$n=0 \Rightarrow$$

$$Q_0 = \alpha_1 3^0 + \alpha_2 (-1)^0 + \frac{5^{0+2}}{12} \quad \text{--- (iii)}$$

$$\Rightarrow 2 = \alpha_1 + \alpha_2 + \frac{25}{12} \quad \text{--- (i)}$$

$$\Rightarrow \alpha_1 + \alpha_2 = 2 - \frac{25}{12} = \frac{1}{12} \quad \text{--- (1)}$$

$$n=1 \Rightarrow$$

$$Q_1 = \alpha_1 3^1 + \alpha_2 (-1)^1 + \frac{5^{1+2}}{12} \quad \text{--- (ii)}$$

$$\Rightarrow 4 = 3\alpha_1 - \alpha_2 + \frac{125}{12} \quad \text{--- (2)}$$

$$\Rightarrow -\frac{77}{12} = 3\alpha_1 - \alpha_2 \quad \text{--- (3)}$$

$$\textcircled{1} - \textcircled{2} \Rightarrow$$

$$\frac{-13}{2} = 4\alpha_1$$

$$\alpha_1 = \frac{-13}{8}$$

$$\alpha_2 = -\frac{1}{12} + \frac{13}{8} \quad \text{--- (ii)}$$

$$\alpha_2 = \frac{37}{24}$$

①, ② → ratio,

$$a_n = -\frac{13}{8} 3^n + \frac{37}{24} (-1)^n + \frac{1}{5} \frac{1^{n+2} T}{12}$$

$$I - \frac{1}{5} \times 10 = \text{Ans}$$

Tower math continuation

$$T_1 = 1$$

$$T_2 = 3$$

$$T_n = 2T_{n-1} + 1 \quad \text{--- (iii)}$$

$$T_n = T_n^h + T_n^p \quad \text{--- (i)}$$

$$T_n^h = 2T_{n-1}$$

$$\Rightarrow r^n = 2r^{n-1} \Rightarrow \boxed{r_1 = 2}$$

$$T_n^h = \alpha_1 r_1^n$$

$$\Rightarrow \boxed{T_n^h = \alpha_1 2^n} \quad \text{--- (ii)}$$

$$T_n^p = A_0 \rightarrow \text{--- (i)}$$

$$1 \rightarrow \text{sin}$$

$$A_0 = 2A_0 + 1$$

$$\boxed{A_0 = -1} \quad \text{--- (iii)}$$

so, no matter you change the n still it is same

21/10

15/10/23

$$a_n = 5a_{n-1} + 84a_{n-2} + 3^n$$

$$a_0 = 2 \text{ (1)}$$

$$a_1 = 8$$

$$\alpha_1 + \alpha_2 = \frac{21}{10}$$

$$12\alpha_1 + \alpha_2 = \frac{81}{10}$$

Number Theory

Divisibility:

Dividend = Divisor $\times$ Quotient + Remainder

Dividend = Divisor $\times$ Quotient + Remainder

$$5 \mid 10 \Rightarrow 5 \mid 10$$

* if, $b \mid a$

$$a = bK \text{ [where } K \in \mathbb{Z}]$$

Problem-1

if $a \mid b$, $b \mid c$ show that, $a \mid c$.

since $a \mid b$,

$$b = aK_1 \text{ [} K_1 \in \mathbb{Z} \text{]}$$

(i)



since $b \mid c$,

$$c = bK_2 \text{ [} K_2 \in \mathbb{Z} \text{]}$$

(ii)

$$b \mid a \Rightarrow a = bK$$

$$a \geq b$$

$$a = bK + r$$

$$17 \geq 5$$

$$\begin{array}{r} 5 \overline{) 17} \\ \underline{15} \\ 2 \end{array}$$

$$17 = 5 \times 3 + 2$$

12/10/23

15

i) $\rightarrow$ ii) = 0

$$c = a k_1 k_2$$

$$\frac{c}{a} = k_1 k_2 = \text{an integer}$$

$\therefore a|c$ [shown]

Q. a/b the show $a|b^2$

$$d \leq d$$

$$a = pk + b$$

$$17 \geq 2$$

$$\begin{array}{r} 17 \\ 2 \overline{) 17} \\ \underline{16} \\ 1 \end{array}$$

$$17 = 2 \times 8 + 1$$

$$a = pk + b \Rightarrow a^2 = p^2 k^2 + 2pb + b^2$$

$$\begin{array}{l} \text{ii) } a = pk + b \\ \text{i) } a = pk + b \end{array}$$

□ is alb and b|c show that, $a^2 | b^2 + ac$ □

since $a|b$,

$$b = ak_1 \quad [k_1 \in \mathbb{Z}]$$

$$\Rightarrow b^2 = a^2 k_1^2 \quad \text{--- (i)}$$

since, $b|c$

$$c = bk_2 \quad [k_2 \in \mathbb{Z}]$$

$$\Rightarrow ac = abk_2 \quad \text{--- (ii)}$$

$$\text{(i)} + \text{(ii)} \Rightarrow$$

$$b^2 + ac = a^2 k_1^2 + abk_2$$

$$= a^2 k_1^2 + a \times (ak_1) \times k_2$$

$$b^2 + ac = a^2 k_1^2 + a^2 k_1 k_2$$

$$b^2 + ac = a^2 (k_1^2 + k_1 k_2)$$

$$\Rightarrow \frac{b^2 + ac}{a^2} = k_1^2 + k_1 k_2 = \text{an integer}$$

so, $a^2 | b^2 + ac$ — (Ans)

Q if $a|b, b|c, c|d$, show that $a|d^3$

$$\begin{aligned}
 & a|b \quad b = ak_1 \quad \text{--- (i)} \\
 & b|c \quad c = bk_2 \quad \text{--- (ii)} \\
 & c|d \quad d = ck_3 \quad \text{--- (iii)}
 \end{aligned}$$

$$\begin{aligned}
 d^3 &= c \times c \times c \times k_3^3 \\
 &= c \times (bk_2) \times (bk_2) \times k_3^3 \\
 &= c \times (bk_2) \times (ak_1 k_2) \times k_3^3
 \end{aligned}$$

$$\Rightarrow d^3 = abc \times k_1 \times k_2^2 \times k_3^3$$

$$\Rightarrow \frac{d^3}{abc} = k_1 \times k_2^2 \times k_3^3 \quad \text{is an integer}$$



$$112 = 1 \times 10^2 + 1 \times 10^1 + 2 \times 10^0$$

$$3746 = 3 \times 10^3 + 7 \times 10^2 + 4 \times 10^1 + 6 \times 10^0$$

$$52 = 5 \times 10^1 + 2 \times 10^0$$

$$10^2 = 99 + 1$$

$$10^3 = 999 + 1$$

$$10^4 = 9999 + 1$$

□ if the sum of the digits of a positive integer is divisible by 3, that positive no. is also divisible by 3.

let,

$$A = A_1 A_2 A_3 A_4 \dots A_n$$

so,

$$3 \mid A_1 A_2 A_3 A_4 + \dots + A_n \text{ then}$$

$$A_1 A_2 A_3 A_4 + \dots + A_n = 3K \quad \text{--- (i) } [K \in \mathbb{Z}]$$

$$A = A_1 A_2 A_3 \dots A_n$$

$$= A_1 \times 10^{n-1} + A_2 \times 10^{n-2} + A_3 \times 10^{n-3} + \dots + A_{n-1} \times 10^1 + A_n \times 10^0$$

$$= A_1 (\underbrace{999 \dots 9}_{n-1} + 1) + A_2 (\underbrace{99 \dots 9}_{n-2} + 1) + A_3 (\underbrace{99 \dots 9}_{n-3} + 1) + \dots$$

$$\dots + A_{n-1} (9 + 1) + A_n$$

$$= A_1 \times 99 \dots 9 + A_1 + A_2 \times 99 \dots 9 + A_2 + A_3 \times 999 + A_3 + \dots + A_{n-1} \times 9 + A_{n-1} + A_n$$

$$= (A_1 + A_2 + A_3 + \dots + A_n) + 3(3333 \dots 3)A_1$$

$$+ 3(33 \dots 3)A_2 + 3(3 \dots 3)A_3 + \dots + 3 \times 3 A_{n-1}$$

$$1 + e + e^2 = 01$$

$$1 + e + e^2 + e^3 = 01$$

$$1 + e + e^2 + e^3 + e^4 = 01$$

$$A = 3K + 3P \quad \left[P = \boxed{\quad} \right]$$

$$A = 3(K+P)$$

$$A = 3 \times (K+P)$$

$$\frac{A}{3} = K+P = \text{an integer}$$

$$\boxed{A} \geq 3 \quad \text{①} \quad - \quad K = \pi A + \dots + \mu A \epsilon A \epsilon A \epsilon A$$

$$\pi A \dots \epsilon A \epsilon A \epsilon A = A$$

$$01 \times_n A + 01 \times_{n-1} A + \dots + 01 \times_\epsilon A + 01 \times_\delta A + 01 \times_1 A =$$

$$\dots + \underbrace{(1+e+\dots+e^{\epsilon-1})}_{\epsilon-n} \epsilon A + \underbrace{(1+e+\dots+e^{\delta-1})}_{\delta-n} \delta A + \underbrace{(1+e+\dots+e^{\mu-1})}_{\mu-n} \mu A =$$

$$\pi A + (1+e) \epsilon A =$$

$$\epsilon A + e \epsilon A + e^2 \epsilon A + \dots + e^{\delta-1} \epsilon A + 1 A + e \dots e \times_1 A =$$

$$\pi A + 1 \pi A + e \times_{n-1} A + \dots +$$

$$1 A (\epsilon \dots \epsilon \epsilon \epsilon \epsilon) \epsilon + (\pi A + \dots + \epsilon A + \delta A + 1 A) =$$

$$\dots + \epsilon A (\epsilon \dots \epsilon \times \epsilon) \epsilon$$

$$1-n A \epsilon \times \epsilon + \dots + \epsilon A (\dots \epsilon) \times \epsilon$$

Prime Power Factorization

* the smallest and the only ^{even} prime number is 2.

$$100 = 2^2 \times 5^2$$

$$126 = 2 \times 63 = 2 \times 7 \times 9 = 2^1 \times 7^1 \times 3^2$$

$$37 = 37^1$$

∴ All natural number can be expressed by the prime factorization.

if $n \in \mathbb{N}$,

$$n = p_1^{\alpha_1} p_2^{\alpha_2} p_3^{\alpha_3} \dots p_{k-1}^{\alpha_{k-1}} p_k^{\alpha_k}$$

□ show that, there are infinite number of primes.

let assume, there are finite number of prime,

$$p_1 p_2 p_3 \dots p_k$$

$$p_1 \times p_2 \times p_3 \times p_4 \dots \times p_k \rightarrow \text{divisible by prime number}$$

$$p_1 \times p_2 \times p_3 \times p_4 \dots \times p_k + 1 \rightarrow \text{divisible by prime number, that means}$$

there is a bigger prime number than we assume p_k

$$8 \times 8 = 8^2 = 64 \quad 9^2 = 81$$

$$\sqrt{64} = 8 \quad \sqrt{81} = 9 \rightarrow 12, 20 \rightarrow \text{GCD}(4)$$

HCF/GCD/ગ.સા.કુ LCM/લ.સા.કુ

Find the GCD (42, 88)

$$42 = 2^1 \times 3^1 \times 7^1$$

$$88 = 2^3 \times 11^1 = 2^1 \times 2^2 \times 11^1$$

$$\text{GCD} = 2^1$$

$$a = 2^3 \times 3^2 \times 5^4 \times 7^9 \times 11^3$$

$$b = 2^5 \times 3^7 \times 5^9 \times 7^2 \times 11^1 \times 13^4$$

$$\text{GCD} = 2^3 \times 3^2 \times 5^4 \times 7^2 \times 11^1$$

(250, 142) $\rightarrow$ GCD

$$250 = 2 \times 125 = 2 \times 5^3$$

$$142 = 2 \times 71$$

$$\text{GCD}(142, 250) = 2$$

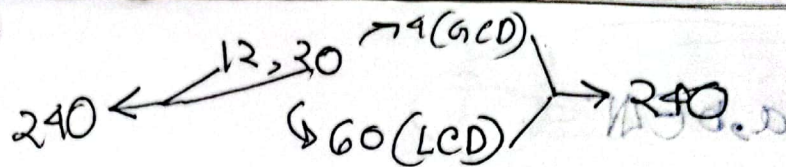
$$a = 2 \times 3 \times 7^3 \times 11$$

$$b = 2 \times 3 \times 7^2 \times 11^2 \times 13$$

$\rightarrow$ L.C.M

$$2^1 \times 3^2 \times 7^3 \times 11^2 \times 13^1$$

MIT
 ↳ 0168863071



(4w) $\boxed{a, b \in \mathbb{N}, a \times b = \text{lcm}(ab) \times \text{gcd}(ab)}$ $\boxed{a \times b = \text{lcm}(ab) \times \text{gcd}(ab)}$

$$\frac{a + (b + d)}{d + d} =$$

$$\frac{a}{d + d} + \frac{(b + d)}{d + d} =$$

$$\frac{a}{d + d} + 1 =$$

$$x = \frac{a}{d + d} + 1$$

$$1 - x = \frac{a}{d + d} \Leftarrow$$

$$a \mid d + d$$

$$x(d + d) = a$$

$$x d + x d = a$$

$$x d = a - x d \Leftarrow$$

$$a - x d = 1$$

$$1 \times d = 1 \times d - 1 \Leftarrow$$

$$1 = d - 1 \rightarrow d = 2$$

$$0 = d - d$$

$$a, b \in \mathbb{N}$$

$$\square \frac{63b + 72a + 15}{8a + 7b} \text{ (is) an integer } a-b=? \square$$

$$= \frac{9(8a + 7b) + 15}{8a + 7b}$$

$$= \frac{9(8a + 7b)}{8a + 7b} + \frac{15}{8a + 7b}$$

$$= 9 + \frac{15}{8a + 7b}$$

$$\text{so, } 9 + \frac{15}{8a + 7b} = k$$

$$\Rightarrow \frac{15}{8a + 7b} = k - 9$$

$$\text{so, } 8a + 7b \mid 15$$

$$15 = (8a + 7b)k$$

$$\Rightarrow 15 = 8ak + 7bk$$

$$\Rightarrow 15 - 8ak = 7bk$$

$$\downarrow$$

$$\textcircled{1} \rightarrow \text{must be}$$

$$\Rightarrow 15 - 8 \times 1 \times 1 = 7b \times 1$$

$$7 = 7b \quad \boxed{b=1} \rightarrow \boxed{a=1}$$

$$\boxed{a-b=0}$$

Q1b

$$\frac{1}{2} \times \frac{2}{1} = 1$$



$$A = A_1 A_2 A_3 \dots A_{n-1} A_n \quad \text{digits not multiplied}$$

now, $4 | A_{n-1} A_n$

$$A_{n-1} A_n = 4K \quad [K \in \mathbb{Z}]$$

$$= A_1 A_2 A_3 \dots A_{n-2} 00 + A_{n-1} A_n$$

$$= 100 \times A_1 A_2 A_3 \dots A_{n-2} + A_{n-1} A_n$$

$$= 100 A_1 A_2 A_3 \dots A_{n-2} + A_{n-1} A_n$$

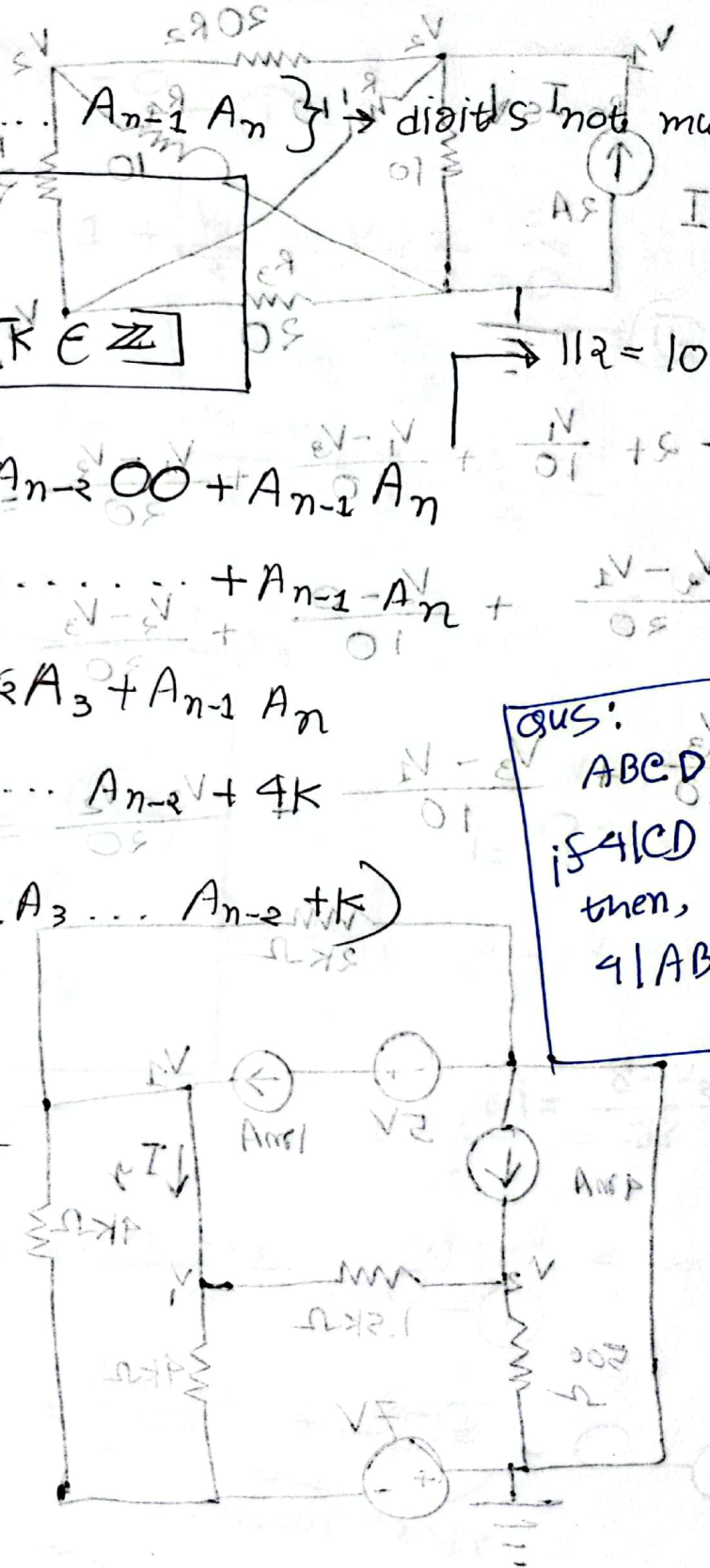
$$= 100 A_1 A_2 A_3 \dots A_{n-2} + 4K$$

$$= 4 (25 A_1 A_2 A_3 \dots A_{n-2} + K)$$

$$A = 4 \times P$$

$$\frac{A}{4} = P = \text{int}$$

$$4 | A \quad \text{if } I \uparrow$$



Qus:
ABCD
is 4 | CD
then,
4 | ABCD

It can solve the
rightmost digit of 7⁹⁹⁷¹

Modular Arithmetic

a) $b \equiv r \pmod{a} \Rightarrow b = ax + r$

$\frac{ax}{r}$ (Divisor) $\rightarrow$ quotient
 $\frac{b}{r}$ (Dividend) $\rightarrow$ remainder (অশেষ)

#

$b \equiv r \pmod{a}$ $\rightarrow$ b is congruent to r of a

Dividend (remainder) Divisor

b যার সাথে a দিয়ে ভাগ দিলে r অবশিষ্ট থাকে

#

5) $12 \pmod{5}$

$\frac{10}{2} \quad 12 \equiv 2 \pmod{5}$

equation এ আমরা
edit করতে পারি,
 $5 = 5$

$$2 + a = b + 2$$

$$2 - a = b - 2$$

$$a + k = b + k$$

$$12 \equiv 2 \pmod{5}$$

$$12 + 1 \equiv 2 + 1$$

$$13 \equiv 3 \pmod{5}$$

$$12 \equiv 2 \pmod{5}$$

$$12 - 1 \equiv 2 - 1$$

$$11 \equiv 1 \pmod{5}$$

* Addition, subtraction
is allowed for
modular Arithmetic

২২/০১/১৯

১৫০০ ১১/০২ মোঃ জে

$$12 \equiv 2 \pmod{5}$$

$$2 \times 12 \equiv 2 \times 2$$

$$24 \equiv 4 \pmod{5}$$

$$\begin{array}{r} 5) 24(4 \\ \underline{20} \\ 4 \end{array}$$

* But division is not always allowed

$$12 \equiv 2 \pmod{5}$$

$$4 \equiv \frac{2}{2} \pmod{5}$$

* Multiplication is also allowed

$$\Rightarrow 6 \equiv 1 \pmod{5}$$

ওমবই ভাগ করা যাবে
যখনই এদের ফলাফল
জোড় রকম না আসবে।

$$12 \equiv 2 \pmod{5}$$

$$12^2 \equiv 2^2$$

$$144 \equiv 4 \pmod{5}$$

* exponent is also allowed

$$12 = 5 \times 2 + 2$$

$$12 = 5 \times 1 + 7$$

$$12 = 5 \times 0 + 12$$

$$12 = 5 \times (-1) + 17$$

$$12 = 5 \times 3 + (-3)$$

Residue

But there is only one on Remainder

$$0 \leq \text{Remainder} < \text{divisor}$$

$$\begin{array}{r} 10 \overline{) 101} \\ \underline{10} \\ 01 \\ \underline{00} \\ 01 \\ \underline{00} \\ 01 \\ \underline{00} \\ 01 \\ \underline{00} \\ 01 \end{array}$$

$$b \equiv r \pmod{a}$$

Residue

$$12 \equiv 2 \pmod{5}$$

$$12 \equiv 7 \pmod{5}$$

$$12 \equiv 12 \pmod{5}$$

$$12 \equiv -3 \pmod{5}$$

You can add/sub 'a' with r that as many times as you want

$$(0 \pmod{10}) + 5 = 5$$

$$(0 \pmod{10}) - 5 = -5$$

$$44 \equiv 4 \pmod{10}$$

$$44 \equiv 14 \pmod{10}$$

$$44 = 10 \times 3 + 4$$

$$44 \equiv 54 \pmod{10}$$

$$= 10 \times (-1) + 54$$

$$44 \equiv 4 \pmod{10}$$

$$44 \equiv -6$$

$$44 = 10 \times 5 + (-6)$$



$$3 \equiv 3 \pmod{8}$$

$$3^2 \equiv 9 \pmod{8}$$

$$3^3 \equiv 1 \pmod{8}$$

$$(3^3)^{48} \equiv 1^{48}$$

$$3^6 \equiv 1 \pmod{8}$$

$$3 \times 3^6 \equiv 3 \times 1$$

$$3^{97} \equiv 3 \pmod{8}$$



(Ans)

$$\begin{array}{r} 10) 52 \text{ (5)} \\ \underline{50} \\ 2 \end{array}$$

$$\begin{array}{r} 10) 127 \text{ (12)} \\ \underline{10} \\ 27 \\ \underline{20} \\ 7 \end{array}$$

what is the rightmost digit of 7^{9997} ?

$$7^{9997} \pmod{10} = ?$$

$$7 \equiv 7 \pmod{10}$$

$$7^2 \equiv 49 \pmod{10}$$

$$7^2 \equiv -1 \pmod{10}$$

$$\Rightarrow (7^2)^{4998} \equiv (-1)^{4998} \pmod{10}$$

$$\Rightarrow 7^{9996} \equiv 1 \pmod{10}$$

$$\Rightarrow 7^{9996} \times 7 \equiv 1 \times 7 \pmod{10}$$

$$\Rightarrow 7^{9997} \equiv 7 \pmod{10}$$

$$\text{just } 7$$

$$7^2 \equiv 49 \pmod{10} \equiv -1 \pmod{10}$$

$$7^4 \equiv 1 \pmod{10}$$

$$(7^4)^{2499}$$

$$(\text{Euler}) 1 = 1 \leftarrow$$

$$(\text{Euler}) 1 = 1 \leftarrow$$

$$\square 3^{47} \equiv ? \pmod{26}$$

$$3 \equiv 3 \pmod{26}$$

$$3^2 \equiv 27 \pmod{26}$$

$$3^3 \equiv 1 \pmod{26}$$

$$(3^3)^5 \equiv 1^{15} \pmod{26}$$

$$\Rightarrow 3^{15} \equiv 1 \pmod{26}$$

$$\Rightarrow 3^{47} \equiv 9 \pmod{26}$$

↑
Remainder.

$$\square 13^{27} \equiv ? \pmod{10}$$

$$13 \equiv 3 \pmod{10}$$

$$13^2 \equiv 3^2 \pmod{10}$$

$$\Rightarrow 13^2 \equiv (-1) \pmod{10}$$

$$\Rightarrow 13^{26} \equiv 1 \pmod{10}$$

$$\Rightarrow 13 \times 13 \equiv -1 \times 13 \pmod{10}$$

$$\Rightarrow 13^{27} \equiv -13 \pmod{10}$$

$$\Rightarrow 13^{27} \equiv 7 \pmod{10}$$

$$\square 7x \equiv 8 \pmod{13}$$

$$x \equiv ? \pmod{13}$$

$$7x \equiv 8 \pmod{13}$$

$$\Rightarrow 7x \equiv 21 \pmod{13}$$

$$\Rightarrow x \equiv 3 \pmod{13}$$

$$\square 11x \equiv 5 \pmod{19}$$

$$x \equiv ? \pmod{19}$$

$$11x \equiv 5 \pmod{19}$$

$$\Rightarrow 11x \equiv 5 - 19 - 19 \pmod{19}$$

$$\Rightarrow 11x \equiv -33 \pmod{19}$$

$$\Rightarrow x \equiv -3 \pmod{19}$$

$$x \equiv 16 \pmod{19}$$

Fermat's little Theorem

$$a^{p-1} \equiv 1 \pmod{p}$$

$$\Rightarrow 3^{7-1} = 1 \pmod{7}$$

$$\Rightarrow 3^6 = 1 \pmod{7}$$

$$\boxed{324}^{715} \equiv ? \pmod{103}$$

$$\text{here } \rightarrow p = 103$$

$$324^{103-1} \equiv 1 \pmod{103}$$

$$\Rightarrow 324^{102} \equiv 1 \pmod{103}$$

$$\Rightarrow (324^{102})^7 \equiv 1^7 \pmod{103}$$

$$\Rightarrow 324^{714} \equiv 1 \pmod{103}$$

$$324^{715} \equiv 324 \pmod{103}$$

$$324^{715} \equiv 15 \pmod{103}$$

24/10/23

$$17x + 3 = 8 \pmod{23}$$

$$x = ? \pmod{23}$$

$$17x + 3 - 3 = 8 - 3 \pmod{23}$$

$$\Rightarrow 17x = 5 \pmod{23}$$

$$17x = 5 + 23 + 23 = 51 \pmod{23}$$

$$\Rightarrow \frac{17x}{17} = \frac{51}{17} \pmod{23}$$

$$x = 3 \pmod{23}$$

Modular Exponentiation

$$15^{69} = ? \pmod{112}$$

$$15^1 \equiv 15 \pmod{112}$$

$$15^2 \equiv 225 - 112 - 112 \pmod{112}$$

$$15^2 \equiv 1 \pmod{112}$$

$$(15^2)^{34} \equiv 1^{34} \pmod{112}$$

$$\Rightarrow 15 \times 15^{68} = 15 \times 1 = 15 \pmod{112}$$

$$15^{69} = 15 \pmod{112}$$

Ans

(*) Any natural number can be expressed as the sum of some powers of 2. যে কোন বৈশিষ্ট্যক পূর্ণসংখ্যাকে ২ এর কয়েকটি ঘাতের যোগফল হিসেবে প্রকাশ করা যায়।

$$1 = 2^0$$

$$\begin{aligned} 230 &= 2^7 + 64 + 32 + 4 + 2 \\ &= 2^7 + 2^6 + 2^5 + 2^2 + 2^1 \end{aligned}$$

$$49 = 2^5 + 2^4 + 2^0$$

$$69 = 64 + 4 + 1$$

Alternative →

$$15^{69} = ? \pmod{112}$$

$$15^{69} = 15^{(64+4+1)}$$

$$15^{69} = 15^{64} \cdot 15^4 \cdot 15^1$$

$$15^{64} = \square \pmod{112}$$

$$15^4 = \square \pmod{112}$$

$$15^1 = \square \pmod{112}$$

$$a \equiv r \pmod{b}$$

$$e \equiv s \pmod{b}$$

$$ae \equiv rs \pmod{b}$$

$$15^1 \equiv 15 \pmod{112} \quad \text{--- (i)}$$

$$15^2 \equiv 225 \pmod{112} = 1 \pmod{112} \quad \text{--- (ii)}$$

$$(15^2)^2 \equiv 1^2 \pmod{112} = 1 \pmod{112} \quad \text{--- (iii)}$$

$$15^4 \equiv 1 \pmod{112} \quad \text{--- (iv)}$$

$$(15^4)^4 \equiv 1^4 \pmod{112}$$

$$15^{64} \equiv 1 \pmod{112} \quad \text{--- (v)}$$

$$\text{(i)} \times \text{(ii)} \times \text{(iii)}$$

$$15^1 \times 15^4 \times 15^{64} = 15 \times 1 \times 1$$

$$116 = 64 + 32 + 16 + 4$$

$$15^{116} \equiv 15 \pmod{112}$$

$$\boxed{\text{Q}} \quad 7^{116} = ? \pmod{107}$$

$$7^{116} = 7^{(64+32+16+4)}$$

$$= 7^{64} \cdot 7^{32} \cdot 7^{16} \cdot 7^4$$

$$7 \equiv 7 \pmod{107}$$

$$7^2 \equiv 49 \pmod{107}$$

$$7^4 \equiv 2401 \pmod{107} = 47 \pmod{107}$$

--- (i)

$$(7 \text{ base}) (7^4)^2 = 47^2 = 2209 \equiv 69 \pmod{107} \quad \text{--- (i)}$$

$$(7 \text{ base}) 7^8 \equiv 69 \pmod{107} \quad \text{--- (ii)}$$

$$(7 \text{ base}) \Rightarrow (7^8)^2 = 4761 \equiv 53 \pmod{107} \quad \text{--- (iii)}$$

$$\Rightarrow 7^{16} \equiv 53 \pmod{107}$$

$$\Rightarrow (7^{16})^2 = 53^2 \pmod{107}$$

$$\therefore 7^{32} = 2809 \equiv 27 \pmod{107} \quad \text{--- (iv)}$$

$$(7^{32})^2 = 27^2 = 729 \pmod{107} = 87 \pmod{107}$$

$$\textcircled{i} \times \textcircled{ii} \times \textcircled{iii} \times \textcircled{iv}$$

$$7^4 \times 7^{16} \times 7^{32} \times 7^{64}$$

$$\equiv 47 \times 53 \times 27 \times 87$$

$$7^{16} = 5851359 \pmod{107}$$

$$= (5851359 - 54685 \times 107)$$

$$7^{16} = 64 \pmod{107}$$

$$(7 \text{ base}) 7^4 = (7 \text{ base}) 1075 = 75$$

1

54685.59813

$\frac{a}{b} = \min$
 $a - b \times m$
 $= \text{min}$

$$\boxed{\text{Q}} \parallel^{57} = ? \pmod{225}$$

$$57 = 32 + 16 + 8 + 1$$

$$= 2^5 + 2^4 + 2^3 + 2^0$$

$$\parallel^{57} = \parallel^{(32+16+8+1)} \pmod{225}$$

$$= \parallel^{32} \cdot \parallel^{16} \cdot \parallel^8 \cdot \parallel^1 \pmod{225}$$

$$\parallel^1 = 11 \pmod{225} \text{--- (i)}$$

$$\parallel^2 = 121 \pmod{225}$$

$$\parallel^4 = 14641 \equiv 16 \pmod{225}$$

$$\parallel^8 = 16^2 = 256 \equiv 31 \pmod{225} \text{--- (ii)}$$

$$\parallel^{16} = 961 \pmod{225} \equiv 61 \pmod{225} \text{--- (iii)}$$

$$\parallel^{32} = \overset{3721}{\cancel{223521}} \pmod{225} \equiv 121 \pmod{225} \text{--- (iv)}$$

$$\text{(i)} \times \text{(ii)} \times \text{(iii)} \times \text{(iv)}$$

$$\parallel^1 \times \parallel^8 \times \parallel^{16} \times \parallel^{32}$$

$$= 11 \times 31 \times 61 \times 121 \pmod{225}$$

$$= 2516921 \pmod{225}$$

$$= 71 \pmod{225}$$

ans

$$\square (256)_6 \times (342)_7 = (x)_8$$

$$\begin{aligned} (256)_6 &= 2 \times 6^2 + 5 \times 6^1 + 6 \times 6^0 \quad (00988) = \\ &= 2 \times 36 + 30 + 6 \\ &= (108)_{10} \end{aligned}$$

$$\begin{aligned} (342)_7 &= 3 \times 7^2 + 4 \times 7^1 + 2 \times 7^0 \\ &= 3 \times 49 + 28 + 2 \\ &= (177)_{10} \end{aligned}$$

$$\begin{aligned} (256)_6 \times (342)_7 &= (108)_{10} \times (177)_{10} \\ &= (19116)_{10} \end{aligned}$$

$$\begin{array}{r}
 9 \overline{) 19116} \\
 \underline{9 2124} \\
 9 \overline{) 236} \\
 \underline{9 26} \\
 9 \overline{) 2} \\
 \underline{9 0}
 \end{array}$$

$$= (28200)_9$$

$$x = 28200$$

$$\square \text{gcd}(a, b) \times \text{lcm}(a, b) = a \times b$$

$$\alpha_1, \alpha_2 \updownarrow \gamma_1, \gamma_2$$

⊛ Prove that if $a, b \in \mathbb{N}$
 let's assume that $\text{gcd}(a, b) = g$; $\text{lcm}(a, b) =$
 since,

$$g|a, g|b$$

$$\Rightarrow a = gk_1$$

$$b = gk_2$$

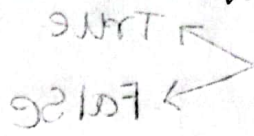
25/11/20

Mathematics

Mathematics

2020-2021

where K_1 and K_2 are relatively prime
(সহজোঁদিক)



$(5, 8) \rightarrow$ সহজোঁদিক

$a|l$ and $b|l$

$$l = an_1 \quad \text{--- (i)}$$

$$l = bn_2 \quad \text{--- (ii)}$$

n_1 and n_2 are co-prime

$$l = an_1$$

$$\frac{l}{a} = n_1$$

$$n_1 = \frac{l}{aK_1} \quad \text{--- (iii)}$$

$$l = bn_2$$

$$\frac{l}{b} = n_2$$

$$\Rightarrow \frac{l}{bK_2} = n_2$$

$$\text{--- (iv)}$$

$$\text{(iii)} \div \text{(iv)} \Rightarrow$$

$$\frac{l \times aK_2}{aK_1 \times l} = \frac{n_1}{n_2}$$

$$\Rightarrow \frac{K_2}{K_1} = \frac{n_1}{n_2} \quad \text{--- (v)}$$

$$\Rightarrow \frac{K_2 \times n_2}{K_1} = n_1$$



৫ ও ৮ দুটি সহজোঁদিক

৫	৮
৮	৫
৫	৮
৮	৫

Propositional Logic

→ Proposition $\begin{cases} \text{True} \\ \text{False} \end{cases}$

$\ast$ denoted with P, q, r, s, a, b

P : Today is Friday $\rightarrow \textcircled{T}$

q : BD is US $\rightarrow \textcircled{F}$

Operator

Negation $\rightarrow \textcircled{\neg P}$

$\neg P$: Today is not Friday

Disjunction

Implication

Conjunction

Biconditional

Exclusive or - Exor

Truth Value

P : BCT is out of ewe2023 $\rightarrow \begin{cases} \textcircled{T} \\ \textcircled{F} \end{cases}$

P	$\neg P$
T	F
F	T

work like (V) of erator

disjunction ($\vee$)

p: Today is Friday

q: It is raining outside

$p \vee q$: Today is Friday or it is raining outside.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Conjunction ($\wedge$)

p: Messi is an Argentine footballer

q: he is a WC winner

$p \wedge q$: p and q

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

যদিও (U) এর ক্ষেত্রে

EX-OR : $\oplus \rightarrow$ one of p and q

হয় p নথুবা q
Either p or q

P: My father's name is Nipu

q: Nasir father's name is Kashem

$P \oplus q$:

P	q	$P \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

Implication : $(P \rightarrow q) \rightarrow$ if p then q

একটি proposition আরেকটির উপর Dependent
P: I get elected

q: I lower the taxes

P	q	$P \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

$P \rightarrow q$: If I do then you do

$P_1 \rightarrow q_1$: If q get elected then I lower the taxes

P	Q	$P \rightarrow Q$
T	T	T
F	T	T
T	F	F
F	F	T

* If P then Q ✓

* If P, Q ✓

* P is sufficient for Q ✓

* P only if Q ✓

$P \rightarrow$ hypothesis ; $Q \rightarrow$ conclusion

Biconditional : $P \leftrightarrow Q$

* P is and only if Q

* P iff Q

* P iff Q

* P is sufficient/necessary and sufficient for Q

for Q $Q \leftarrow P$

$\neg P$

T

F

F

F

$\neg Q$

T

F

T

F

$P \leftrightarrow Q$

T

F

F

T

T

F

T

F

T

T

F

F

$\textcircled{*}$ Tautology $\rightarrow$ যদি কোন proposition এর truth values always T, then that proposition is a tautology
 $\textcircled{*}$ contradiction

<u>P</u>	<u>$\neg P$</u>	<u>$P \vee \neg P$</u>
T	F	T
F	T	T

$(P \vee \neg P) \rightarrow \text{Tautology}$

what is the truth table for the proposition
 $(P \wedge q) \oplus (q \rightarrow P)$

P	q	$P \wedge q$	$q \rightarrow P$	$(P \wedge q) \oplus (q \rightarrow P)$
T	T	T	T	F
T	F	F	T	T
F	T	F	F	F
F	F	F	T	T

Precedence of logical operators

$$\neg > \wedge > \vee > \oplus > \rightarrow > \leftrightarrow$$

Logical Equivalence

output and if the order of (T and F) same then we can say these two proposition logical equivalence

$$(P \leftrightarrow Q) \vee (Q \oplus P)$$

□ $P \leftrightarrow Q$ and $\neg(Q \oplus P)$ is logically equivalent

$P \leftrightarrow Q$	$\neg(Q \oplus P)$
T	T
F	F
F	F
T	T

$$\frac{51}{0}$$

$\square 24^{29} \equiv ? \pmod{29} / 24^{29} \pmod{29} = ?$

$24^{28} \equiv 1 \pmod{29}$ [Fermat's theory]

$24^{29} \equiv 24 \pmod{29}$

Euclidean Algorithm

48, 126

$48 \overline{) 126} \quad (2)$
 $\underline{96}$

$30 \overline{) 48} \quad (1)$
 $\underline{30}$

$18 \overline{) 30} \quad (1)$
 $\underline{18}$

$12 \overline{) 18} \quad (1)$
 $\underline{12}$

$6 \overline{) 12} \quad (2)$
 $\underline{12}$
 0

G.C.D

$$\square \text{ GCD}(72, 156)$$

$$\begin{array}{r} 72 \overline{) 156} \quad 2 \\ \underline{144} \\ 12 \quad 72 \quad 6 \\ \underline{72} \\ 0 \end{array}$$

GCD

$$\text{GCD} = 12$$

$$72 \times 156 = 12 \times \text{LCM}$$

$$\boxed{\text{LCM} = 936}$$

$$\text{GCD} \times \text{LCM} = a \times b$$

$$a \equiv 0 \pmod{b}$$

$$a \equiv 0 \pmod{b}$$

$$a \equiv 1 \pmod{b}$$

$$a \equiv 1 \pmod{b}$$

$$\square 20x \equiv 7 \pmod{123}$$

$$\square \text{gcd}(p, q) = d_1$$

$$\text{gcd}(r, s) = d_2$$

$$\text{lcm}(r, s) = l_1$$

$$\text{lcm}(p, q) = l_2$$

$$p \times q = d_1 \times l_2$$

$$r \times s = d_2 \times l_1$$

$$p \times r \times q \times s = d_1 d_2 \times l_1 l_2$$

$$q^2 = \frac{d_1 d_2 \times l_1 l_2}{pr} \Rightarrow q = \sqrt{\frac{d_1 d_2 l_2}{m}}$$

$$11 \times 5^{20n} = 11^{8n+3}$$

$$5 \equiv 0 \pmod{5}$$

$$5^{20n} \equiv 0^{20n} \equiv 0 \pmod{5}$$

$$11 \equiv 1 \pmod{5}$$

$$11^{8n+3} \equiv 1^{8n+3} \pmod{5}$$

$$\equiv 1 \pmod{5}$$

$$9 \equiv 4 \pmod{5}$$

$$5^{20n} + 11^{8n+3} + 9$$

$$\equiv 0 + 1 + 4 \pmod{5}$$

$$\equiv 0 \pmod{5}$$

$$1 \times 10 = 1 \times 9$$

$$1 \times 10 = 9 \times 1$$

$$1 \times 10 = 9 \times 1$$

$$\frac{1 \times 10}{1} = \frac{9 \times 1}{1}$$

$$(203) (155) (100) (100)$$

$$04$$

$$05 (5)$$

$$06$$

$$05 (5)$$

$$05$$

$$0$$

$$(100)$$

$$100 = 100$$

$$5 \times 120 = 600$$

$$5 \equiv 2 \pmod{3}$$

$$5 \equiv -1 \pmod{3}$$

$$5^{20n} \equiv (-1)^{20n}$$

$$11 \equiv 2 \pmod{3}$$

$$11 \equiv 1 \pmod{3}$$

$$11^{8n+3} \equiv 1^{8n+3} \pmod{3}$$

$$\equiv 1 \pmod{3}$$

$$\equiv 1 \pmod{5}$$

$$\boxed{\square} (55^{400} - 55^{200})$$

$$55 \equiv 1 \pmod{3}$$

$$55^{400} \equiv 1^{400} \pmod{3}$$

$$\equiv 1 \pmod{3}$$

$$55^{200} \equiv 1 \pmod{3}$$

$$55^{400} - 55^{200} \equiv 1 - 1 \equiv 0$$

Decimal $\leftrightarrow$ All

$$\boxed{\square} P(n) \quad 1+2+3+4+\dots+n = \frac{n(n+1)}{2}$$

$$Q(n) \div 1+2+3+\dots+n = \frac{n(n+1)}{2}$$

Step 1 = Base Step

Step 2 + Step 3 = Inductive Step

$$\boxed{\square} P(n) : 1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Step-1 $n=1$

$$L.H.S = 1 = 1$$

$$R.H.S = 1$$

$$n = (m-1)$$

$$(m-1) \times m \times \dots \times (m-1) \times 1$$

Implication $\rightarrow$

$$p \rightarrow q$$

$$\text{Converse: } q \rightarrow p$$

$$\text{Inverse: } \neg p \rightarrow \neg q$$

$$\text{Contrapositive: } \neg q \rightarrow \neg p$$

p : It rains

q : BD wins

*only if

BD wins, it rains

If it doesn't rain then BD doesn't win

If BD doesn't win then it doesn't rain

quantifier

Existential

$\exists$

universal

$\forall$

For all,

$P(x)$: x is a perfect square

For all x , $P(x)$

$P(x)$: $x > 3$

$\forall x P(x)$

There exists some x , $P(x)$

propositional function

→ propositional logic

$\exists x P(x)$

propositional function

$P(x)$: x is greater than 2

propositional variable

$P(2)$: False

$P(3)$: True

$P(x)$: x contains an 'a'

$P(\text{orange})$: True

$P(\text{purple})$: False

$P(x)$: $x^2 + 2$ is odd

$P(3)$: $9 \Rightarrow$ True

domain = set of all english words

→ universe of discourse

$\phi(x): x$ is greater than 5

[domain is the set of real numbers]

$\exists x \phi(x)$ is a Truth value,

$\rightarrow$ True

$\forall x \neg \phi(x) \rightarrow x$ is not greater than 5

$\rightarrow$ False True False

$\neg \forall x \phi(x) \rightarrow x$ is greater than 5

$\rightarrow$ False

$\phi(x): x$ is a politician

$R(x): x$ is honest

Domain: is the set of all humans.

$\forall x \neg \phi(x) \rightarrow$ False

~~$$\exists x \phi(x) = \neg \forall x \neg \phi(x)$$~~

~~$$\forall x \neg \phi(x) = \neg \exists x \phi(x)$$~~

$$\textcircled{*} \exists x \neg \phi(x) = \neg \forall x \phi(x)$$

$\phi(x)$: x is a good human

$$\exists x \neg \phi(x) = \neg \forall x \phi(x)$$

$$\forall x \neg \phi(x) = \neg \exists x \phi(x)$$

□ All lions are fierce, some fierce creatures don't drink coffee, some lion drink coffee.

Domain = set of all creatures.

$P(x)$: x is fierce

$\phi(x)$: x is a lion

$R(x)$: x drinks coffee

$$\forall x (\phi(x) \rightarrow P(x)) \rightarrow \neg (\exists x (\phi(x) \wedge \neg P(x)))$$

$$\exists x (P(x) \wedge \neg R(x))$$

$$\exists x (\phi(x) \wedge R(x))$$



↳ ना use करी
जाएगी

$\square$ No woman is happy, some happy women do not cook. Some women who cook are educated. All educated woman are not happy.

$P(x)$: x is happy

domain: women

$Q(x)$: x cooks

$R(x)$: x is educated

~~$\forall x (P(x) \rightarrow \neg Q(x))$~~

$\exists x (P(x) \wedge \neg Q(x))$

$\exists x (Q(x) \wedge R(x))$

$\forall x (R(x) \rightarrow \neg P(x))$

$((\exists x (P(x) \wedge \neg Q(x)) \wedge \exists x (Q(x) \wedge R(x)) \wedge \forall x (R(x) \rightarrow \neg P(x))) \rightarrow \neg \forall x (P(x) \rightarrow \neg Q(x)))$

$\exists x (P(x) \wedge \neg Q(x))$

$\exists x (Q(x) \wedge R(x))$

$\leftarrow x \in$

3/10/2020